

Worksheet
Integration: Substitution / Integration by Parts

1. $\int \frac{dx}{x \ln x}$

2. $\int_0^2 \frac{e^x}{1+e^x} dx$

3. $\int \frac{dy}{\sqrt{y}(1+y)}$

4. $\int_0^{\pi/2} \frac{\sin \theta}{\sqrt{1+\cos \theta}} d\theta$

5. $\int \frac{e^x}{1+e^{2x}} dx$

6. $\int_0^4 x\sqrt{2x+1} dx$

7. $\int u^3 \sqrt{u^2+9} du$

8. $\int_1^2 \frac{e^{1/x}}{x^2} dx$

9. $\int x \cos 2x dx$

10. $\int_0^1 t^2 e^t dt$

11. $\int \tan^{-1} x dx$

12. $\int_0^1 \ln(x^2+1) dx$

13. $\int \frac{z^5}{\sqrt{1+z^2}} dz$

14. $\int_0^{3/4} \frac{s}{\sqrt{1-s}} ds$

15. $\int \frac{\sin 3x}{\cos^5 3x} dx$

16. $\int_0^{16} \frac{dx}{x^{1/4}+1}$